CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-501: Introduction to DBMS

Session: December 2024 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) What is Schema in DBMS?

Definition: A **Database Schema** is the formal structural blueprint and design description of the entire database. It specifies:

- Table names, columns, data types, and domains.
- Integrity constraints (primary keys, foreign keys, check constraints).
- Relationships between entities and storage representations.

It is stored in the system catalog and does not change frequently.

Q2 (3M) What are the advantages of using DBMS?

Key Advantages:

1. **Minimal Redundancy:** Normalization ensures data is stored in one primary location.
2. **Data Integrity:** Automatic enforcement of domain and referential integrity constraints.
3. **Concurrency Control:** Multi-user simultaneous access without collisions via locking protocols.
4. **Crash Recovery:** WAL (Write-Ahead Logging) and checkpoints guarantee durability after hardware failure.
5. **Data Independence:** Separates user views from low-level storage modifications.

Q3 (3M) Who is a DBA? Mention their key functions.

Definition: A **Database Administrator (DBA)** is an IT professional who manages the database environment, maintains performance, controls user permissions, schedules backups, and enforces security policies.

Q4 (3M) What is a Database?

Definition: A **Database** is an integrated and logically related collection of records, designed to meet the information processing needs of multiple users across an organization.

Q5 (3M) What is RDBMS?

Definition: A **Relational Database Management System (RDBMS)** is a DBMS based on E.F. Codd's Relational Model (1970). Data is organized into 2D tables (relations) consisting of rows (tuples) and columns (attributes), manipulated using SQL and relational algebra. Examples: Oracle, MySQL, PostgreSQL, SQL Server.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Find the minimum number of tables required for the given ER diagram in the relational model. Explain table conversion rules.**

Rules for ER to Relational Reduction:

1. **Strong Entity Sets:** Each strong entity converts into a separate independent table.
2. **Weak Entity Sets:** Converts into a table containing its discriminator (partial key) combined with the primary key of the identifying strong entity as composite primary key.
3. **1:1 Relationships:** Combine both entities into 1 table, or place the primary key of one table as a foreign key in the other table.
4. **1:N Relationships:** Place the primary key of the '1' side entity into the table of the 'N' side entity as a foreign key (no new table needed).
5. **M:N Relationships:** Requires a new separate relationship table containing primary keys of both participating entities as foreign keys.

Result: For an ER model with two strong entities A and B in 1:N relationship, exactly **2 tables** are needed. If the relationship is M:N, exactly **3 tables** are required.

Q7 (7.5M) **What are UNION, MINUS, and INTERSECT commands in DBMS? Explain with syntax and examples.**

Conditions: Both queries must be *Union-Compatible* (same number of columns with compatible data types).

- `UNION` : Combines results of two SELECT queries and eliminates duplicate tuples.
- `INTERSECT` : Returns only the common tuples present in both result sets.
- `MINUS / EXCEPT` : Returns tuples from the first query that are not present in the second query.

SQL Examples:

```
-- UNION: All students from CS or IT
SELECT RollNo, Name FROM Student_CS
UNION
SELECT RollNo, Name FROM Student_IT;

-- INTERSECT: Students enrolled in both Math and Physics clubs
SELECT StudentID FROM MathClub
INTERSECT
SELECT StudentID FROM PhysicsClub;

-- MINUS: Registered students who have NOT paid exam fees
SELECT StudentID FROM RegisteredStudents
EXCEPT
SELECT StudentID FROM PaidStudents;
```

Q8 (7.5M) **Define the following terms: (a) Data Model, (b) Database Schema.**

(a) Data Model: A conceptual collection of tools, constructs, and notations used to describe data, data relationships, data semantics, and consistency constraints. Major models include:

- **Relational Model:** Tables of tuples and attributes (SQL).
- **Entity-Relationship (ER) Model:** High-level conceptual modeling.
- **Object-Oriented & Document (NoSQL) Models:** JSON, key-value, graph.

(b) Database Schema: The overall structural skeleton of the database defined during the design phase, distinct from the dynamic instances/states stored inside it.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Define Normal Forms. Explain 1NF, 2NF, 3NF, and BCNF with suitable examples.**

Detailed Normal Forms Analysis:

Normal Form	Requirement / Condition	Anomaly Eliminated
1NF	All attribute values must be atomic (no multivalued / nested attributes).	Repeating groups and unstructured data arrays
2NF	In 1NF and no non-prime attribute is partially dependent on any candidate key.	Partial Functional Dependencies
3NF	In 2NF and for every X → Y, either X is a Super Key or Y is a Prime Attribute.	Transitive Functional Dependencies
BCNF	For every non-trivial functional dependency X → Y, X must strictly be a Super Key.	All functional dependency based redundancies

BCNF Example:
Suppose `STUDENT_ADVISOR(StudentID, Subject, Advisor)` where each student has an advisor per subject, and each advisor teaches only one subject: FD_1: (StudentID, Subject) \to Advisor, FD_2: Advisor \to Subject.
Candidate Key is (StudentID, Subject) or (StudentID, Advisor).
In FD_2, `Advisor` is not a super key, violating BCNF.
Decompose into: `Advisor_Subject(Advisor, Subject)` and `Student_Advisor(StudentID, Advisor)`. Both are now in BCNF!

Q10 (15M) **Explain Normalization with its advantages. What anomalies arise in unnormalized databases?**

Anomalies in Unnormalized Tables:

1. **Insertion Anomaly:** Cannot record information about an entity without having to insert another unrelated entity (e.g., cannot register a new department until at least one employee is hired).
2. **Deletion Anomaly:** Deleting a record of one entity unintentionally erases information about another entity (e.g., deleting the last employee in a department deletes all department details).
3. **Update / Modification Anomaly:** Changing an entity's attribute requires updating hundreds of duplicate rows; if one row is missed, data becomes contradictory.

Advantages of Normalization:

- Eliminates data redundancy and conserves disk storage.
- Maintains referential integrity automatically across tables.
- Speeds up row updates, inserts, and index lookups.

Q11 (15M) **Explain SQL with its various characteristics and features.**

Structured Query Language (SQL): The ANSI/ISO standard declarative language for relational databases.
Key Characteristics:

- **Declarative / Non-Procedural:** Users specify *what data* they want, leaving the query optimizer to determine *how* to retrieve it.
- **Comprehensive Capabilities:** Encompasses DDL (structure), DML (manipulation), DCL (security/privileges), and TCL (transactions).
- **Portability & Standards:** Works uniformly across major vendors (Oracle, MySQL, SQL Server, PostgreSQL).
- **Integrity Constraints Support:** Native enforcement of `PRIMARY KEY`, `FOREIGN KEY`, `CHECK`, `UNIQUE`, and `DEFAULT`.
- **View & Subquery Support:** Enables nested queries, common table expressions (CTEs), and dynamic virtual tables (Views).

Q12 (15M) **What is Relational Algebra? Discuss its basic and extended operations with expressions.**

Overview: Relational algebra provides the formal theoretical mathematical foundation for query evaluation and execution plan optimization in RDBMS.
Operations Summary:

- **Selection (σ):** Horizontal subset of rows based on conditions.
- **Projection (π):** Vertical subset of columns, duplicate-free.
- **Cartesian Product (×):** All pairs of concatenated rows.
- **Set Union (∪), Intersection (∩), Difference (−):** Set-theoretic operations on union-compatible relations.
- **Theta Join ($\bowtie_\theta$):** Cross product followed by condition filtering.
- **Division (÷):** Used for queries involving "for all" / universal quantification.

Q13 (15M) **Explain the reduction of ER Schema to Relational Tables with a complete case study.**

Case Study: Company Database:
Entities: `EMPLOYEE`, `DEPARTMENT`, `PROJECT`.
Relationships:

- `WORKS_FOR` (N:1 from Employee to Department)
- `MANAGES` (1:1 from Employee to Department with `StartDate`)
- `WORKS_ON` (M:N between Employee and Project with `Hours`)

Relational Reduction:

```
1. DEPARTMENT(DeptNo, DeptName, MgrEmpID, MgrStartDate)
   -- 1:1 MANAGES folded into DEPARTMENT with MgrEmpID as FK
2. EMPLOYEE(EmpID, EmpName, Salary, DeptNo)
   -- N:1 WORKS_FOR represented via DeptNo as FK referencing DEPARTMENT
3. PROJECT(ProjNo, ProjName, DeptNo)
   -- N:1 CONTROLS represented via DeptNo as FK
4. WORKS_ON(EmpID, ProjNo, Hours)
   -- M:N relationship becomes a separate junction table with composite PK
   -- FK EmpID → EMPLOYEE, FK ProjNo → PROJECT
```
